

SILVA NURZANATUL DAHMALENA

Informatics Engineering Student | AI/ML Engineer Enthusiast

silvadahmalena14@gmail.com | 083813042280 | Github | Portofolio | LinkedIn

About Me

Informatics Engineering student with hands-on experience in developing AI and machine learning applications through academic projects and practical training. Skilled in Python, Pandas, NumPy, and Scikit-learn, with experience in building RAG-based chatbots, integrating LLMs through APIs, and developing interactive AI applications. Currently participating in the AI for Business Independent Study Program at NF Academy. Passionate about developing practical AI solutions and exploring machine learning technologies.

Education

STT TERPADU NURUL FIKRI — **2024 - Present**

Bachelor of Informatics Engineering | GPA: 3.90/4.00

NF ACADEMY — **2026 - Present**

Independent Study Program – AI for Business

Technical Skills

- **Programming:** Python, SQL
- **Data Processing:** Pandas, NumPy, Data Cleaning, Data Validation, Data Preprocessing, EDA, Feature Engineering
- **Machine Learning & AI:** Scikit-learn, NLP, Model Evaluation, LLM Integration, RAG, FAISS
- **API & Backend:** API Integration, REST API
- **AI Tools & Frameworks:** Groq API, Google Gemini API, LangChain, n8n
- **Database & Tools:** MySQL, Supabase, Microsoft Excel, Google Sheets, Streamlit, Git

Project Experience

ACADEMIC INFORMATION CHATBOT | RAG-BASED AI SYSTEM — **Mei-June 2026**

- Collected academic information from campus websites and academic posters through web scraping using BeautifulSoup and text extraction using Tesseract OCR.
- Performed text preprocessing and chunking to prepare unstructured academic data for retrieval.
- Implemented semantic retrieval using multilingual-e5-base embeddings and FAISS vector indexing.
- Integrated Llama 3.3 through Groq API to generate context-aware responses based on retrieved academic information.
- Collaborated with a four-member team to test and refine the chatbot's retrieval pipeline.

- **SKRIPSIBOT | RAG-BASED ACADEMIC WRITING CHATBOT** **September 2026**
  - Designed and deployed a web-based AI assistant to support academic writing and final-project guidance using PDF reference documents.
  - Developed an automated workflow using n8n by integrating Google Gemini, Supabase Vector Store, and chat memory.
  - Connected document retrieval from Google Drive to generate context-aware responses grounded in academic references.
  - Conducted end-to-end testing through the n8n workflow and published web interface.

- **NEWS CATEGORY CLASSIFICATION | MACHINE LEARNING & NLP** **Mei-June2026**
  - Processed and analyzed more than 24,000 labeled news articles across multiple categories using text preprocessing and feature extraction techniques.
  - Developed and compared classification algorithms, including Naive Bayes, Logistic Regression, and Support Vector Machine (SVM), using TF-IDF and Word2Vec.
  - Achieved 96.8% classification accuracy using SVM with TF-IDF and developed an interactive Gradio application to demonstrate the model.

- **BOGOR TRAFFIC VOLUME PREDICTION | MACHINE LEARNING & DATA ANALYTICS** **April-June 2026**
  - Collected, cleaned, and processed historical traffic data using Python and Pandas to prepare datasets for analysis and modeling.
  - Performed Exploratory Data Analysis (EDA) and feature engineering to identify patterns and relevant factors influencing traffic volume.
  - Developed and evaluated machine learning models using Scikit-learn to predict traffic volume.
  - Built an interactive Streamlit dashboard to visualize prediction results and present data insights.

Organization Experience

- **DPM (STUDENT REPRESENTATIVE COUNCIL)** **2026-Present**
  - Collected, understood, and documented student feedback, concerns, and needs regarding campus services.
  - Managed student aspirations and coordinated with student organizations and campus stakeholders to support effective follow-up.
  - Supported student service initiatives through collaborative communication and documentation.

- **GWD (GROWTH WITH DATA)** **2026**
  - Participated in collaborative learning activities and discussions during data-related case studies.
  - Worked with team members to identify problems, evaluate possible solutions, and apply structured problem-solving approaches.

- **BEM (STUDENT EXECUTIVE BOARD)** **2025-2026**
  - Supported student advocacy and service programs by collecting and communicating student concerns.
  - Assisted in coordinating activities focused on improving student services and responding to student needs.

Committe Experience

- **ORMIK – ORIENTATION PROGRAM | MENTOR** **2026**
  - Mentored approximately 50 new students and provided guidance to help them adapt to academic and campus life.
  - Facilitated group discussions and encouraged active participation among mentees.

- **INTERNATIONAL CAMPUS TALK | SECRETARY** **2025**
  - Managed administrative documents, meeting minutes, and event documentation.
  - Coordinated communication between committee divisions and tracked follow-up actions.
  - Assisted in scheduling meetings and monitoring follow-up actions to support event preparation.
- **INFORMATICS ENGINEERING ORIENTATION PROGRAM | EVENT COMMITTEE** **2025**
  - Managed administrative documents, meeting minutes, and event documentation.
  - Coordinated communication between committee divisions and tracked follow-up actions.
  - Collaborated with committee members to address operational and technical needs while facilitating participant engagement.